

STUDENT ENGINEERING PROJECT · SEPTEMBER 2026

AQI TOWER

From an Idea to an Engineering Foundation

A student-led exploration of air-cleaning architecture, computational simulation, filtration science, and responsible product development — built on evidence, bounded by honesty.

PHASES COMPLETED

18+

CFD SIMULATION CASES

10

PHYSICAL PROTOTYPES BUILT

0

FAKE RESULTS INCLUDED

0

CONCEPT VISUAL — DIGITAL MODEL ONLY
No physical prototype exists

What if a simple idea about cleaner air could be explored like a real engineering system?

The AQI Tower project began with a visual concept: a vertical tower-shaped machine that moves indoor air through multiple treatment stages to deliver cleaner air. Rather than immediately building something and hoping it worked, the team chose a different path.

AI tools helped organize research, inspect technical documents, automate workflows, and accelerate visualization. OpenFOAM simulated airflow in computational detail. Real manufacturer data replaced estimated numbers. Each result — including uncomfortable ones — changed the design.

This report tells that story honestly: what was built, what was discovered, what was changed as a result, and what remains to be done by experienced professionals and physical testing.

"Simulation is most valuable when it changes the design."

— Core principle of this project

Air contains many different types of pollutants.

"AQI" — the Air Quality Index — is a number that reflects several different pollutants simultaneously. No single technology removes all of them, and different technologies target different particle sizes and chemical species.

Understanding this distinction drove every major architecture decision in this project.

Fine Particles (PM_{2.5})

Particles smaller than 2.5 μm . Penetrate deep into lungs. Require HEPA-grade mechanical filtration. Cannot be removed by activated carbon alone.

H13 HEPA TARGET

Gases & VOCs (e.g. HCHO)

Molecules like formaldehyde pass straight through HEPA filters unchanged. Require adsorption media (activated carbon) or chemistry-specific treatment.

CARBON STAGE RESEARCH TARGET

Coarse Dust (PM₁₀+)

Larger particles settle quickly. A washable panel pre-filter handles these efficiently, protecting fine filters from rapid clogging.

PHASE 18: G4/M5 PRE-FILTER RECOMMENDED

Why this matters: Every architecture decision — filter type, treatment sequence, carbon media, fan power — follows from this pollutant physics. Getting it wrong produces a device that looks credible but does not address the health-relevant fraction.

A vertical tower. Multiple treatment stages. One continuous airflow.

The concept is straightforward: ambient air enters one side, passes through a sequence of treatment stages driven by a fan, and cleaner air exits from the top.

The original treatment sequence proposed in concept documents included particle filtration, activated carbon for gas-phase treatment, an experimental water-based stage, sensors, and eventually solar-assisted power. These were hypotheses, not validated choices.

TREATMENT CONCEPT STAGES

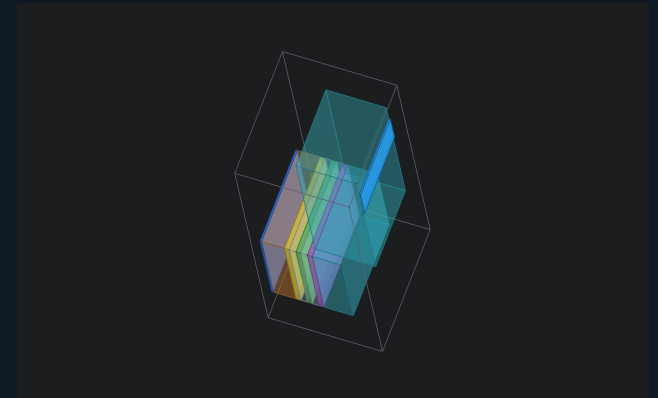
- Inlet → dirty-air plenum
- Pre-filter (coarse particles)
- H13 HEPA (fine particles)
- Activated carbon (gases)
- Clean riser → KVO 250 fan
- Outlet

STATUS OF WATER STAGE

Originally proposed as an experimental wet-separation stage. Phase 18 (2026) conducted a full engineering feasibility review.

PHASE 18 RESULT: NO-GO

Water spray replaced by G4/M5 dry pre-filter — better physics, no hygiene risk.



CONCEPT VISUAL — DIGITAL MODEL ONLY

Simulation before fabrication. Evidence before claims.

The team made an early commitment: every design claim would be traceable to a real source — manufacturer data, peer-reviewed literature, a computational simulation result, or a planned physical experiment. Invented numbers were not accepted.

This discipline slowed some decisions but produced a project where every number can be followed back to its origin, and every uncertainty is visible.

- ⦿ **Requirements & concept design**
Application, treatment hypothesis, architecture exploration
- ⦿ **Parametric CAD (FreeCAD)**
Identifiable airflow regions, not a fixed object
- ⦿ **Airflow CFD (OpenFOAM)**
V00, V01: real fan curve, RANS simulation
- ⦿ **Geometry optimisation**
GEO_A / B / C — systematic comparison
- ⦿ **Filter + bypass research**
HEPA model, bypass discovery, sealing fix
- ⦿ **Particle tracking study**
4 sizes, 224 seeds each, frozen RANS field
- ⦿ **Carbon / gas research & experiment design**
HCHO target frozen; packed-bed rig fully specified
- ⦿ **Water-spray feasibility review (Phase 18)**
NO-GO — replaced by dry pre-filter
- ⦿ **Physical experiments + expert review**
Next required stage — not yet performed

✓ AI helped us

- + Organise, read, and cross-reference technical documents and research papers
- + Generate, debug, and audit CFD configuration scripts and Python analysis pipelines
- + Extract fan-curve data from manufacturer PDFs using vector-path analysis
- + Identify engineering questions and decision branches the team had not yet considered
- + Compare alternatives with structured decision matrices
- + Build the interactive 3D engineering viewer (Three.js, T1–T3)
- + Document every assumption, limitation, and evidence-quality classification
- + Accelerate the pace of structured engineering exploration

✗ AI did not replace

- Physical experiments — no airflow, pressure, or gas measurements have been made
- Expert engineering judgement in HVAC, filtration, or industrial product development
- Safety review, regulatory analysis, or certification strategy
- Manufacturing knowledge — enclosure design, tolerances, assembly, sealing
- Laboratory validation — no CADR, no PM removal efficiency, no breakthrough curve
- Professional accountability — results are engineering-student outputs, not certified designs

"AI made exploration faster. It did not make untested results true."

Every tool in this project has a specific, traceable role.

FreeCAD

Parametric 3D CAD software used to create and modify the tower geometry. All dimensions traceable to named parameters, not fixed coordinates.

OPEN SOURCE

OpenFOAM

Open-source computational fluid dynamics solver. Used for all airflow simulations — velocity, pressure, turbulence — inside the tower geometry.

ALL CFD RESULTS FROM THIS

ParaView

Scientific visualisation software. Post-processes OpenFOAM output to produce velocity/pressure field images and extract quantitative data.

ALL CFD IMAGES VIA PARAVIEW

Python

Analysis and automation scripting. Used for data processing, Darcy–Forchheimer fitting, particle tracking, visualisation scripts, and export pipelines.

SCRIPTS IN REPOSITORY

Three.js

Browser-based 3D visualisation library. Powers the interactive engineering viewer (T1–T3) displaying geometry, particles, and CFD scalar fields in a web browser.

VIEWER INCLUDED IN REPOSITORY

Manufacturer Data

Real product specifications from Systemair (KVO 250 fan curve), Freudenberg (H13 HEPA filter data), Sensirion (sensor specs), and Kuraray/Calgon Carbon (media data).

SOURCED, NOT ESTIMATED

Academic Literature

Peer-reviewed aerosol science, filtration engineering, adsorption theory, and air quality standards. Used to bound engineering claims and identify failure modes.

EVIDENCE-CLASSIFIED THROUGHOUT

AI Assistants

Used as a research accelerator, code reviewer, and documentation assistant. All engineering claims verified against primary sources — AI output was never accepted unchecked.

RESEARCH TOOL — NOT A SOURCE

Git Version Control

All project files, simulation cases, scripts, data, and documents tracked in a single repository. Every change logged. Every number traceable.

FULL AUDIT TRAIL

Instead of drawing one fixed object, the team developed an adjustable parametric architecture.

A parametric CAD model means every dimension — the riser width, filter face area, chamber depths, turning geometry — is a named, adjustable variable rather than a fixed number. Changing one dimension propagates correctly throughout the model.

This made it practical to run systematic geometry comparisons later in the project without rebuilding from scratch.

TOWER AIRFLOW ZONES

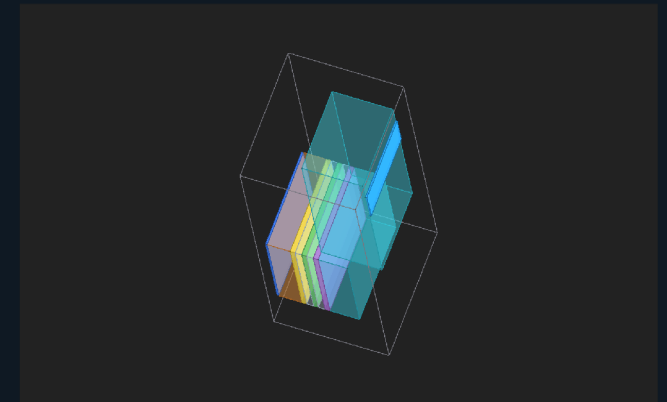
- AIR INLET — horizontal entry
- DIRTY PLENUM — flow distribution
- STAGE REGIONS — treatment cassettes
- CLEAN PLENUM — air collection
- TURNING CHAMBER — direction change
- CLEAN RISER — vertical rise to fan
- OUTLET — fan discharge

GEO_C KEY DIMENSIONS

Riser inner width: **260 mm**
Riser top height: **1850 mm**
Filter face: **593 × 593 mm**
Two-step turning chamber

GEO_C BEST CURRENT GEOMETRY

Dimensions from parametric CAD audit — not yet final design



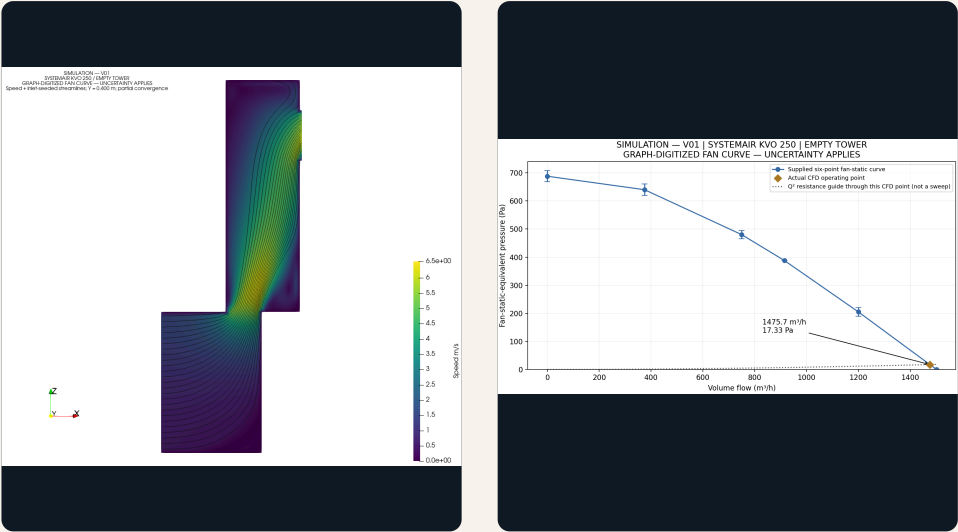
DIGITAL MODEL — FreeCAD Parametric
No physical prototype exists

A good-looking airflow path is not necessarily a good airflow path.

The first airflow simulations (V00, placeholder 1 m/s inlet) revealed that the original tower geometry produced strong recirculation in the riser, uneven velocity distribution, and pressure losses that would reduce the fan's working margin significantly.

Phase V01 introduced the actual Systemair KVO 250 fan performance curve — six data points from the manufacturer catalogue. Instead of assuming a fixed inlet velocity, the simulation now found its own operating point where fan output matched system resistance.

Key finding from V01: At the real fan operating point, the empty tower produced 17.33 Pa of total pressure loss — modest but real. The riser showed 23.1% reverse-flow by volume, indicating significant recirculation that would impair performance.



Left: V01 velocity field. Right: Fan curve intersection — where system resistance meets fan output.

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

1,476

M³/H V01
AIRFLOW

SIMULATION ONLY

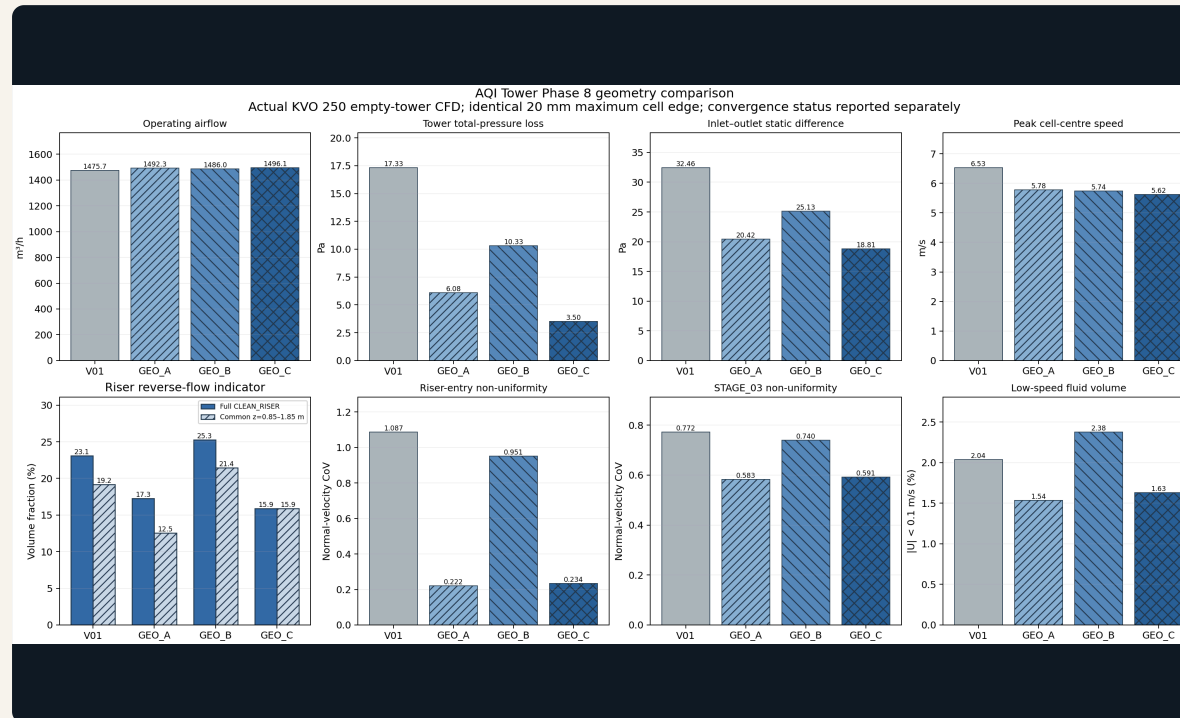
17.3
Pa

V01 PRESSURE
LOSS
SIMULATION ONLY

23%

RISER REVERSE-
FLOW
SIMULATION ONLY

Could we make the air path significantly easier for the fan?



Phase 8 geometry comparison — V01, GEO_A, GEO_B, GEO_C velocity and pressure fields.

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

V01

17.3 Pa

pressure loss
23.1% riser reverse

GEO_B

10.3 Pa

pressure loss
25.3% riser reverse
REJECTED — worse recirculation

GEO_A

6.1 Pa

pressure loss
17.3% riser reverse
PROMISING — did not converge

GEO_C ★

3.5 Pa

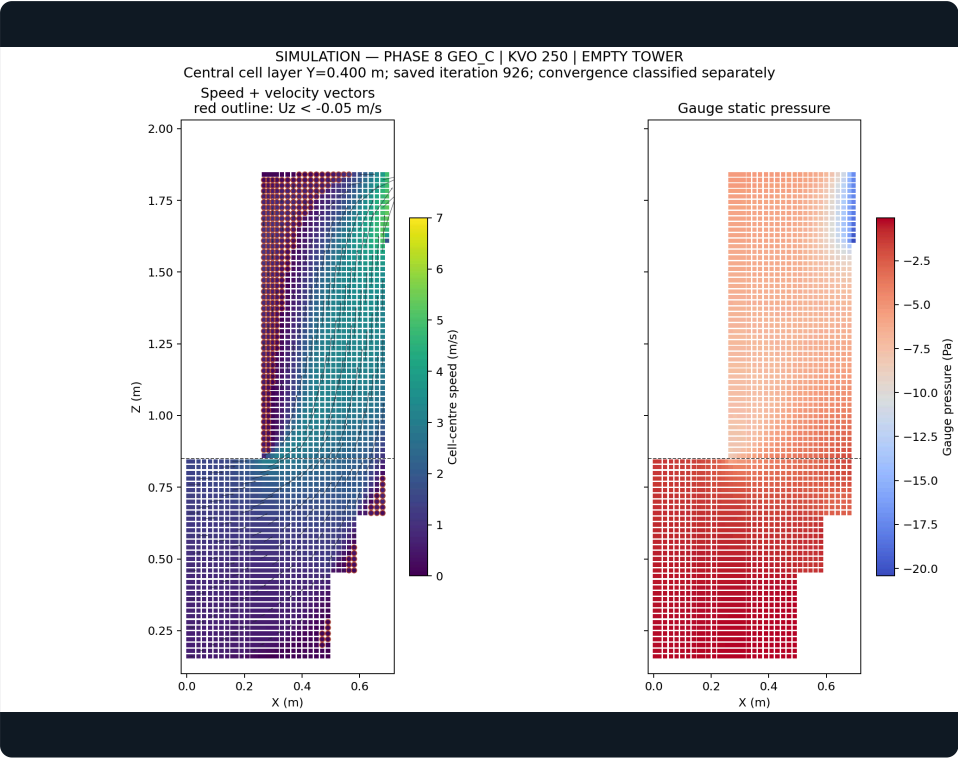
pressure loss
15.9% riser reverse
BEST CURRENT — CONVERGED

GEO_C: the critical improvement

Pressure loss reduced **5× from V01 to GEO_C** (17.3 → 3.5 Pa).
Riser entry uniformity improved dramatically (CoV 1.09 → 0.23). GEO_C confirmed deterministic reproducibility — run twice, identical results.

SIMULATION ONLY

PHYSICAL VALIDATION NOT YET DONE



GEO_C velocity magnitude and pressure fields — centre-plane longitudinal section.

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

1,496 m³/h

Simulated airflow — GEO_C empty tower

SIMULATION ONLY. Fan: Systemair KVO 250. No filters. Not measured.

3.5 Pa
FAN HEAD (EMPTY)

63,780
COMPUTATIONAL CELLS

926
ITERATIONS TO CONVERGE

0.23
RISER ENTRY COV (V01: 1.09)

Systemair KVO 250 Fan

6-point curve from manufacturer catalogue. Shutoff head: **~688 Pa ± 20 Pa**.
Curve-digitized via PyMuPDF vector extraction.

A fan is not a constant-flow device. As system resistance increases (adding filters, media beds), the fan delivers less airflow. GEO_C's low empty-tower resistance preserves headroom for filtration stages.

MANUFACTURER DATA — 6 CURVE POINTS CONFIRMED

NOT FINAL FAN SELECTION

H13 HEPA — what the classification actually means

The project selected the **Freudenberg SF13-B-0593×0593×292** as the research candidate HEPA filter. At 593 × 593 × 292 mm, it matches the tower cross-section with a gross face area of 0.351 m².

H13 is a classification under EN 1822. It means the filter achieves ≥99.95% efficiency at the *Most Penetrating Particle Size* (MPPS) — the one size that is hardest to capture. For glass fibre HEPA, MPPS is typically 0.1–0.25 µm.

What H13 does not automatically mean

An H13 classification on the filter component does not directly translate to 99.95% PM2.5 removal for the complete tower. System performance depends on: bypass leakage paths, face velocity differences from test conditions, frame sealing, and airflow distribution uniformity.

Particle-size efficiency data for this specific product are not publicly available. The EN 1822 test certificate accompanying each purchased unit contains the actual fractional efficiency curve and MPPS — this document must be obtained on purchase.

FILTER SPECIFICATION

Freudenberg SF13-B

0593 × 0593 × 292 mm

H13 per EN 1822

V12x25-N10N-J27-ACA

MANUFACTURER CONFIRMED

CONFIRMED DATA

≥99.95% at MPPS

300 Pa at 2.84 m/s (clean)

Rated flow: 3600 m³/h

MANUFACTURER DATA

NOT YET AVAILABLE

Efficiency at 0.3, 1.0, 2.5, 10 µm

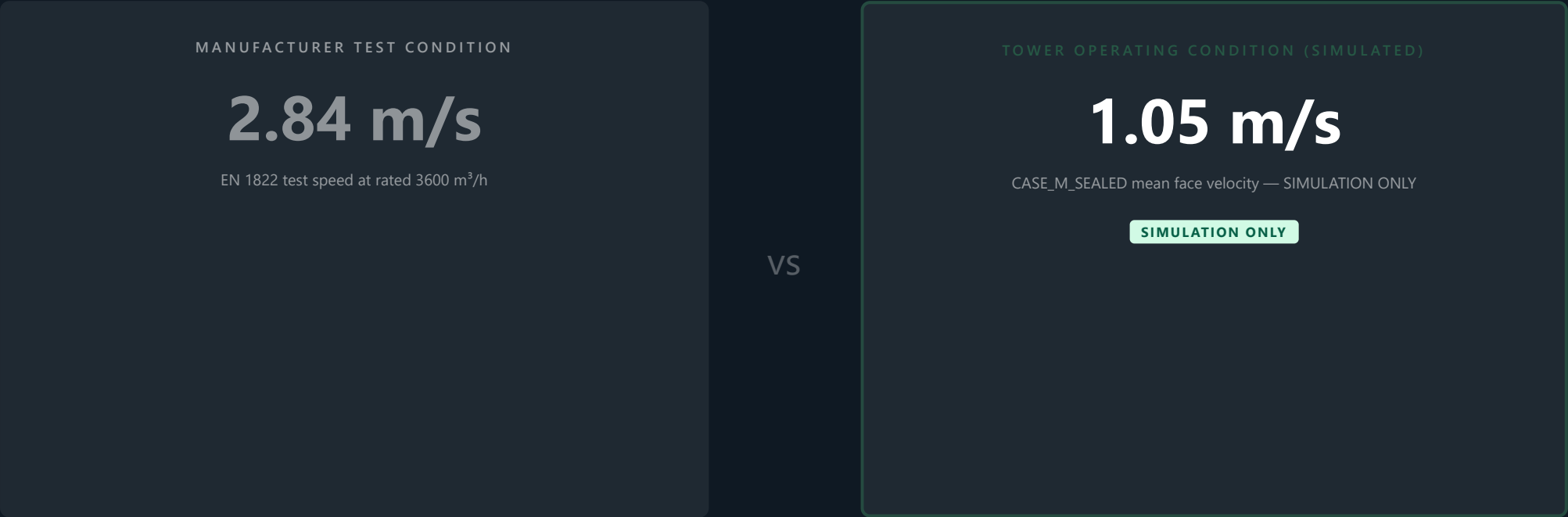
MPPS particle diameter

ΔP vs velocity curve

India purchase price

NOT YET OBTAINED

The filter must be evaluated at the tower's operating face velocity — not at the manufacturer's rated test conditions.



The tower operates at 1.05 m/s — less than 37% of the manufacturer's test velocity. At lower face velocities, diffusion efficiency for ultrafine particles may increase, but inertial impaction and interception change. The exact effect at this product's MPPS is not publicly available — it is in the EN 1822 certificate issued with each unit.

In the sealed-filter simulation

Airflow: **1,332 m³/h**
Filter ΔP: **111 Pa** (vs 300 Pa at 2.84 m/s)
Face velocity CoV: **0.10** — reasonably uniform

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

What this means for the design

Lower face velocity reduces filter ΔP — leaving more fan headroom. But it also means the filter certificate data must be checked at 1.05 m/s, not 2.84 m/s, to confirm efficiency bounds remain valid.

The simulation found something we had not intended to design.

When the first filter model was added to the GEO_C geometry, the simulation revealed that **45% of all simulated airflow was bypassing the filter face** — travelling around the filter through an unintended open path at the junction between the dirty plenum and the clean riser.

This was not an assumed simplification. The path was geometrically real in the model. Unfiltered air reached the clean side of the system.

The team investigated the geometry, identified the exact bypass location ($z = 0.850$ m, $x = 0.260$ – 0.305 m overlap zone), designed a corrective baffle — the *bypassSeal* plate — and reran the simulation.

After sealing — CASE_M_SEALED

45%

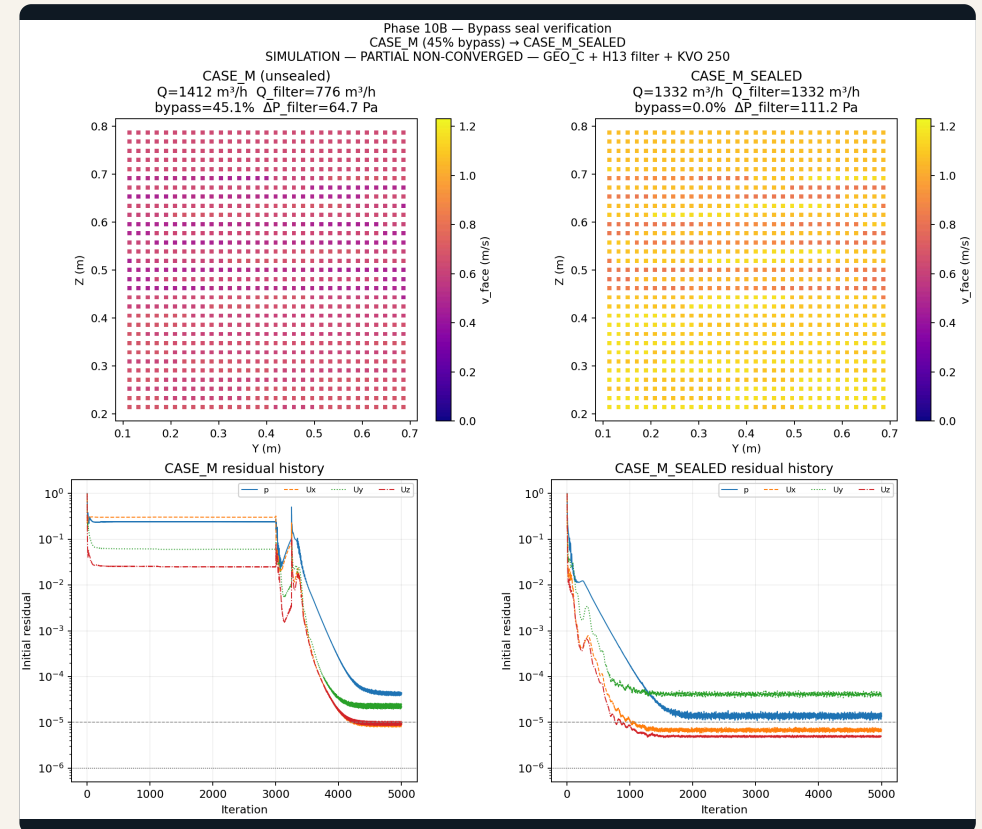
bypass BEFORE



~0%

bypass AFTER (modelled)

CFD bypass $\approx 0\%$ means the simulated geometry no longer contains the modelled bypass path. It does **not** prove a fabricated tower will be physically leak-free — sealing quality depends on manufacturing and assembly.



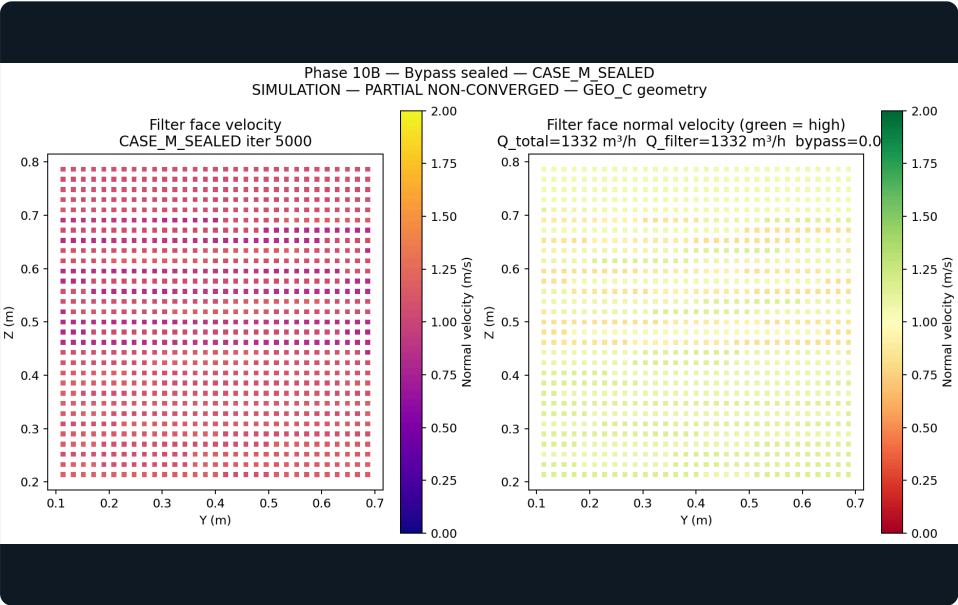
Bypass streamlines before and after the bypassSeal baffle.

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

1,332

m³/h — simulated airflow with H13 HEPA

SIMULATION ONLY — NOT PHYSICALLY VALIDATED



HEPA filter face velocity distribution — CASE_M_SEALED.
Mean 1.053 m/s, CoV 0.10 — simulated, not measured.

What the sealed simulation shows

111 Pa

FILTER ΔP
simulation

1.053 m/s

MEAN FACE VELOCITY
simulation

~0%

BYPASS FRACTION
modelled — not measured

5,000

ITERATIONS RUN
partially converged RANS

Important limitations

The filter model is calibrated to a single manufacturer data point (300 Pa at 2.84 m/s) — one point is insufficient for a validated pressure-drop model. RANS simulation is partially converged. The filter is modelled as a uniform porous baffle, not a real pleated media structure. All results are screening-level estimates only.

We released simulated particles into a frozen airflow field and followed where they travelled.

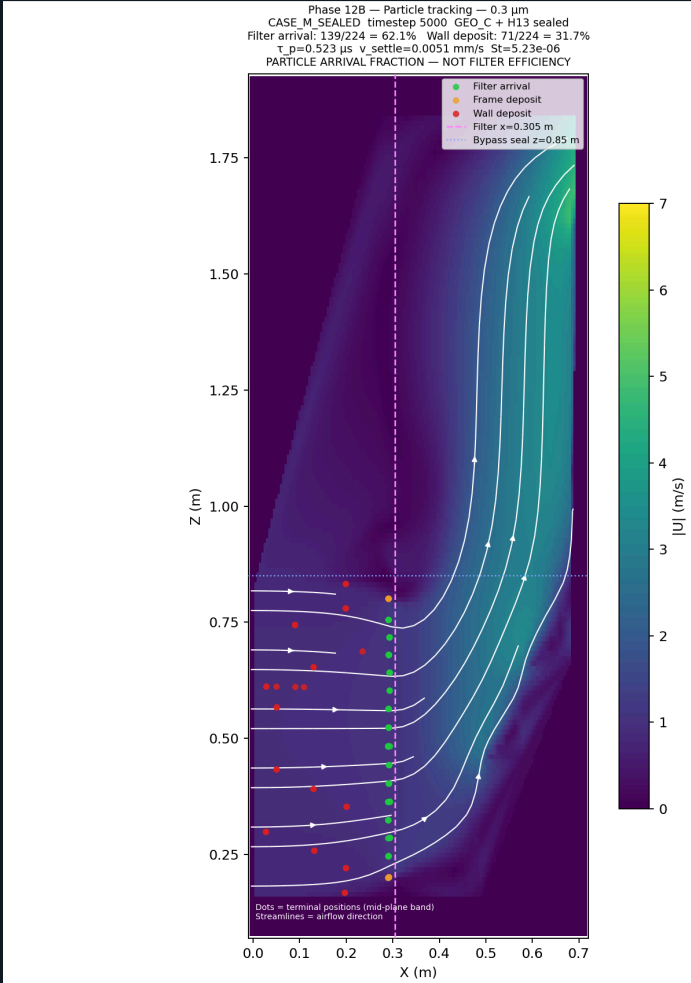
Phase 12B injected 224 seed particles per size class at the tower inlet — representing particles of 0.3, 1.0, 2.5, and 10 μm diameter — and tracked their paths through the CASE_M_SEALED velocity field using an equilibrium-slip Euler model.

The model includes gravity but not turbulent dispersion. It is a conservative, simplified model — not a full Lagrangian simulation with stochastic turbulence. Results show where particles tend to travel, not where every real particle will go.

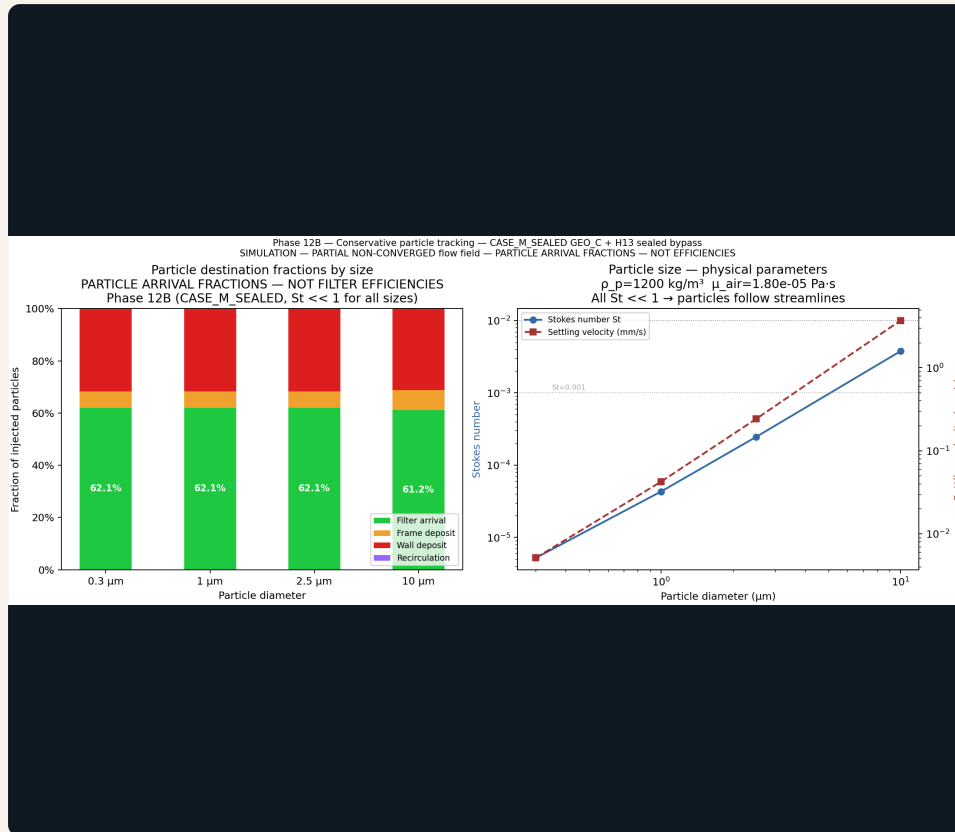
Results per size class (224 seeds each)

0.3 MM (PM2.5 PROXY)	1.0 MM
139/224 → filter face (62.1%)	139/224 → filter face (62.1%)
85/224 → walls / frame (37.9%)	85/224 → walls / frame (37.9%)
2.5 MM	10 MM
139/224 → filter face (62.1%)	137/224 → filter face (61.2%)
85/224 → walls / frame (37.9%)	87/224 → walls / frame (38.8%)

SIMULATION ONLY **ARRIVAL ≠ CAPTURE**



Simulated 0.3 μm particle paths — CASE_M_SEALED frozen field.
SIMULATION ONLY — NOT PHYSICALLY VALIDATED



Phase 12B summary — all four particle sizes.

SIMULATION ONLY — NOT PHYSICALLY VALIDATED

FILTER ARRIVAL $\neq$ FILTER CAPTURE

62% of simulated particles reaching the filter face means exactly that: 62% arrived at the plane where the filter is located. It does not mean 62% were captured, and it says nothing about what percentage of particles were removed from the air.

Filter capture efficiency — the fraction actually retained in the filter media — depends on the product's fractional efficiency curve $\eta(d_p)$. This data is not yet available for the Freudenberg SF13-B (Phase 12A finding). Without $\eta(d_p)$, no PM removal efficiency can be calculated.

Similarly, the 37–39% of particles that contacted walls or frame — this is a model output about simulated contact events. Whether real particles would genuinely deposit there, or re-entrain into airflow, depends on particle composition, surface energy, and airflow fluctuations not modelled here.

Why this matters: Oversimplifying particle tracking results into a CADR or efficiency claim would be technically indefensible. The particle study is most useful for understanding geometry, flow topology, and identifying regions of the design that deserve investigation.

HEPA filters are invisible to gas molecules. A completely separate treatment mechanism is needed.

Formaldehyde (HCHO), benzene, toluene, and other volatile organic compounds (VOCs) are molecular in size — several orders of magnitude smaller than the particles that HEPA captures. They pass through HEPA media unchanged.

Gas-phase removal requires **adsorption** — the process by which molecules bind to the surface of a porous solid medium. Activated carbon is the most established material for this purpose.

After evaluating multiple gas species (formaldehyde, benzene, toluene, acetone, ethanol, generic TVOC), Phase 14 selected **formaldehyde (HCHO)** as the primary research target. It ranked first in a documented weighted decision matrix on health significance, indoor prevalence, measurability, and research tractability.

The primary carbon candidate is **Calgon Carbon OVC 4×8** — treated as a non-impregnated research control. A secondary candidate (FORMASORB) is a manufacturer-claimed HCHO-specific impregnated carbon.

RESEARCH TARGET GAS

HCHO

Formaldehyde — CAS 50-00-0

Confirmed Phase 14 decision

RESEARCH TARGET

PLANNED TEST CONDITIONS

100 µg/m³ baseline HCHO

50 / 200 µg/m³ comparison

25 ± 2 °C, 50 ± 5% RH

High-humidity stress: 70% RH

NOT YET MEASURED

CURRENT CARBON EVIDENCE

OVC 4×8: manufacturer pressure graph only

No HCHO breakthrough data

No Darcy–Forchheimer coefficients

EXPERIMENT PENDING

Sometimes the correct engineering decision is not to simulate yet.

To model how a packed carbon bed behaves in CFD, two things are essential: a pressure-drop coefficient describing how the media resists airflow (Darcy–Forchheimer coefficients a and b), and ideally some mass-transfer data describing how quickly HCHO adsorbs.

Neither set of data exists yet for OVC 4×8 at the conditions relevant to this tower. The manufacturer publishes a pressure graph, but it does not cover the velocity ranges this tower uses, and no HCHO mass-transfer coefficient has been published for this specific grade at these conditions.

Rather than inventing coefficients, the project designed a physical experiment to measure them. This is why gas-stage CFD is blocked: **blocked for the right reason.**

The gas-stage CFD model will be built once real measured inputs are available. Running it with invented numbers would produce a credible-looking but meaningless result.

GAS CFD STATUS

BLOCKED

Waiting for measured inputs from Phase 17A experiment

WHAT IS READY

- ✓ Experiment protocol defined
- ✓ Analysis pipeline coded
- ✓ CFD model structure planned
- ✓ Data schema and templates ready

WHAT IS MISSING

- X Pressure-drop measurements
- X HCHO breakthrough curves
- X Physical hardware

Before putting carbon into the full tower — measure it in a smaller, controlled column.

The bench-scale test rig measures the pressure drop of a packed OVC 4×8 carbon bed across multiple bed depths and airflow velocities. This produces the Darcy–Forchheimer coefficients needed for CFD. A subsequent experiment measures HCHO breakthrough curves.

Column geometry

- 75 mm internal diameter
- Bed depths: 26 / 52 / 104 mm
- 5 velocity points per depth
- 3 replicates per condition

NOT YET BUILT

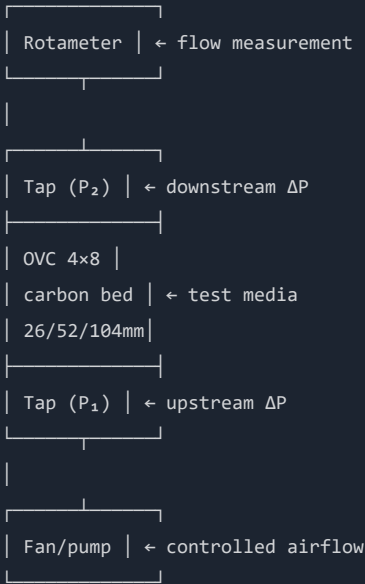
Key instruments

- Sensirion SDP810 (0–500 Pa, 1 Pa res.)
- Rotameter 0–100 L/min
- SHT40 temperature/humidity
- Raspberry Pi 4B data logger

INSTRUMENTS SPECIFIED

Procurement status: SDP810 confirmed at ₹4,748 (Tanotis India). Rotameter A confirmed at ₹1,950. Carbon media (OVC 4×8) requires quote from Kuraray/Chemet India. HCHO testing requires institutional safety approval. No items ordered yet.

CLEAN-AIR PRESSURE-DROP RIG SCHEMATIC



Bottom-to-top flow. $\Delta P = P_1 - P_2$.
SDP810 across taps. PLANNED — NOT BUILT.

Ready in method. Waiting for physical execution.

Hardware procurement has been intentionally deferred. The institutional safety approval required for HCHO generation has not yet been obtained. These are gating conditions, not design failures.

All analysis templates, Python pipelines, data schemas, procurement lists, and validation procedures exist and are fully specified. The experiment can begin as soon as hardware is in hand and safety approval is granted.

No experimental values have been fabricated. All pressure-drop fields, Darcy–Forchheimer coefficients, and breakthrough metrics are explicitly recorded as null or NOT MEASURED until measurements are performed.

✓ Complete and ready

Phase 15–17 experiment protocol
Column geometry and BOM
Python analysis pipeline
Data schema (14 mandatory fields)
India vendor procurement plan
RFQ letters for OVC 4×8 / FORMASORB
CSV template (headers only — no data)

II Blocked pending

OVC 4×8 availability and price quote
Acrylic column fabrication (75 mm ID)
Institutional safety approval (HCHO)
External HPLC laboratory agreement

The original concept included a water-spray stage. Phase 18 reviewed it rigorously. The result was NO-GO.

Phase 18 was a research-only engineering feasibility assessment — no CFD, no physical experiment. The analysis reviewed aerosol physics, pressure-drop impact, biological risk standards, water handling complexity, and alternatives.

The core physics problem: simple low-energy spray towers are ineffective for PM_{2.5} (0.1–2 µm). This is the "Greenfield gap" — the particle size range where neither inertial impaction nor diffusion produces useful collection in a compact chamber.

Why the water stage was rejected

- **PM_{2.5} irrelevant:** Greenfield gap — simple spray cannot capture 0.3–2.5 µm particles at this scale
- **Coarse dust better served** by dry G4/M5 pre-filter — same function, no water, no risk
- **Hygiene risk is disqualifying:** Legionella and biofilm in recirculating warm-water spray (ASHRAE 188, CIBSE TM13)
- **Carryover threatens HEPA:** droplets reaching filter face cause wetting and microbial loading
- **35–125 Pa added ΔP** — reduces airflow 7–23% from an already-constrained operating point
- **12+ extra components** — pump, tank, nozzles, demister — for no unique benefit

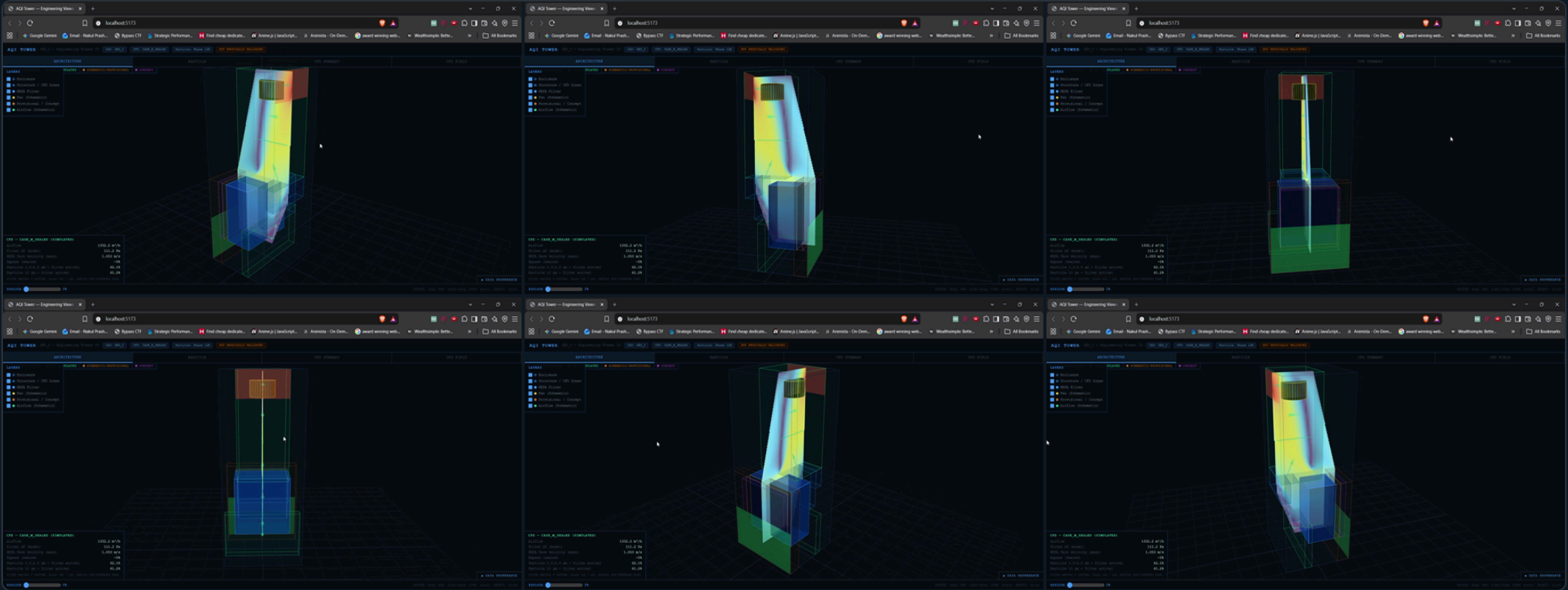
Replacement: G4/M5 washable pre-filter

Achieves coarse pre-collection upstream of HEPA. No water, no pump, no Legionella risk. One component. ~25–50 Pa clean ΔP. Washable monthly. Widely available in India at low cost.

New treatment sequence: Inlet → [G4/M5 washable pre-filter] → [Activated carbon bed] → [H13 HEPA] → Clean riser → KVO 250 fan → Outlet

A browser-based 3D viewer built with Vite + Three.js — displays geometry, particle tracks, and real CFD scalar fields directly in any web browser, no software install required.

Vite + Three.js + ES modules · localhost:5173



T1 — ARCHITECTURE

Interactive 3D geometry. Click any component for engineering details. Layer toggles. Exploded view. Status labels.

T2 — PARTICLE MODE

Phase 12B particle tracking. Filter by size 0.3–10 µm. FILTER ARRIVAL ≠ CAPTURE always visible.

T3 — CFD FIELD

Real OpenFOAM velocity + pressure scalar fields. Turbo colourmap. Convergence badge. Labelled SIMULATION ONLY.

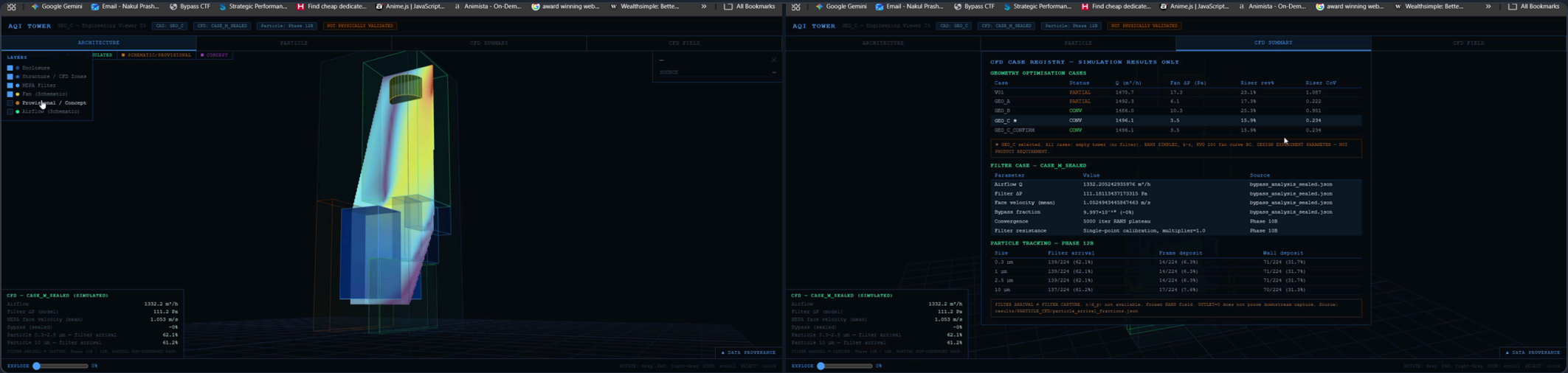
CFD SUMMARY

Exact JSON-sourced values: 1,332 m³/h, 111 Pa. Provenance panel per case. Labelled PARTIAL NON-CONVERGED.

VIEWER ENFORCES

- ✓ Claims labelled by source
- ✓ ARRIVAL ≠ CAPTURE in particle mode
- ✓ Convergence status per CFD case
- ✓ Provenance: path, timestep, derivation
- ✓ Viewer does not create engineering truth

Stills captured from the live Three.js viewer session. Left: CFD scalar field mode — velocity magnitude across GEO_C geometry. Right: CFD Summary panel — exact simulation values pulled directly from solved JSON files.



CFD FIELD VIEW (LEFT)

Velocity magnitude coloured by turbo colourmap. CASE_M_SEALED geometry. Real OpenFOAM output, exported as scalar planes and rendered in browser. **SIMULATION ONLY — not physically validated.**

CFD SUMMARY PANEL (RIGHT)

Geometry optimisation case register + CASE_M_SEALED filter + particle tracking — all values read live from JSON. Expandable provenance links to source file paths. **Results are SIMULATED — no physical prototype tested.**

Video recordings available: The full interactive viewer session has been recorded. Videos show all four modes (Architecture, Particle, CFD Field, CFD Summary), 3D rotation, layer toggles, and provenance panels in real time. Repository: [GIF/main three js.mp4 \(17 s\)](#) · [GIF/20260907-0914-13.1778659.mp4 \(73 s\)](#)

What exists today — and what does not

Developed and documented

- Project requirements framework and architecture concepts
- Parametric FreeCAD tower model (GEO_C)
- Real fan data — KVO 250, 6-point curve, 688 Pa shutoff
- HEPA filter research — Freudenberg H13, evidence gaps identified
- Airflow CFD — 10 cases from V00 to CASE_M_SEALED
- Geometry optimisation — GEO_C: 5× pressure reduction vs V01
- Bypass discovery and fix — bypassSeal baffle in simulation
- Phase 12B particle tracking — 4 sizes, 224 seeds, arrival fractions
- Formaldehyde research target confirmed (Phase 14)
- Carbon media selection — OVC 4×8 as research control
- Bench-test rig engineering (Phases 15–17): protocol, BOM, scripts
- Water spray reviewed and rejected (Phase 18) — G4/M5 recommended
- Interactive 3D engineering viewer (Three.js T1–T3)
- Procurement research — India vendor routes, confirmed prices

Not yet performed or validated

- Physical airflow measurement in any tower geometry
- Physical pressure drop across any real component
- Clean Air Delivery Rate (CADR) — not established at any level
- PM removal efficiency — at any particle size
- Carbon packed-bed pressure drop measurements
- HCHO breakthrough curve for OVC 4×8
- Filter fractional efficiency $\eta(d_p)$ for Freudenberg product
- Physical enclosure sealing performance
- Energy consumption / power budget
- Gas-phase CFD (blocked — no measured inputs)
- Solar-assisted power (CONCEPT status only)
- Any physical prototype or manufactured unit

01

A good-looking airflow path is not necessarily a good airflow path.

V00 had plausible visual flow but significant recirculation and pressure losses. Three geometry iterations were needed to reach a well-behaved field.

03

Filters create large pressure penalties that shift the operating point.

The H13 HEPA added ~111 Pa — reducing airflow from ~1,496 to ~1,332 m³/h. Every treatment stage competes for the same fan headroom.

05

Particle trajectories are not filtration efficiency.

62% of simulated particles reaching the filter face is a geometry-routing result, not a removal efficiency. No capture probability was applied because it is not yet available.

07

Missing data should lead to an experiment — not an invented coefficient.

Gas-phase CFD was blocked, correctly. The packed-bed experiment exists specifically to generate the inputs that will make CFD meaningful.

02

A fan must be evaluated with system resistance.

Adding real fan curve data (V01) fundamentally changed the simulation. A fan delivers the flow at which system resistance equals fan output — not a fixed number.

04

Bypass paths matter as much as filter efficiency.

45% bypass — discovered in simulation — showed that a highly rated filter can be functionally irrelevant if air routes around it. Finding this early was the value of the simulation.

06

Gas removal is pollutant-specific and requires pollutant-specific evidence.

A documented comparison of six gas species was needed before HCHO could be selected. Generic "carbon removes VOCs" is not an engineering claim.

08

A stage that looks useful can be eliminated by physics.

The water-spray stage was part of the original concept. Phase 18 showed that basic aerosol physics makes it ineffective for PM_{2.5} while introducing serious hygiene risks. Removing it was correct.

From exploration to professional engineering.

The project has produced a structured engineering foundation: real simulation results, real component data, identified failure modes, and explicit documentation of every uncertainty. The next maturity level requires experienced people.

The existing project gives domain professionals something concrete to review — not a blank sheet. Questions can be answered within a defined context rather than starting from nothing.

The result of this project is not a finished purifier.
The result is a better-defined engineering problem.

DOMAIN EXPERTISE NEEDED AT NEXT STAGE

HVAC & Airflow

Architecture, fan selection, duct design, real pressure drop

Filtration Engineering

Filter mounting, sealing, bypass prevention in manufactured enclosure

Mechanical Design

Tolerances, assembly, materials, structural loads, maintenance access

Adsorption / Chemistry

Carbon sizing, humidity effects, bed life, HCHO specifics

Laboratory Testing

Protocol design, DNPH reference method, measurement chain

Electrical & Controls

Power budget, motor control, sensor integration

Safety & Regulatory

Chemical safety, IP ratings, certification path in India

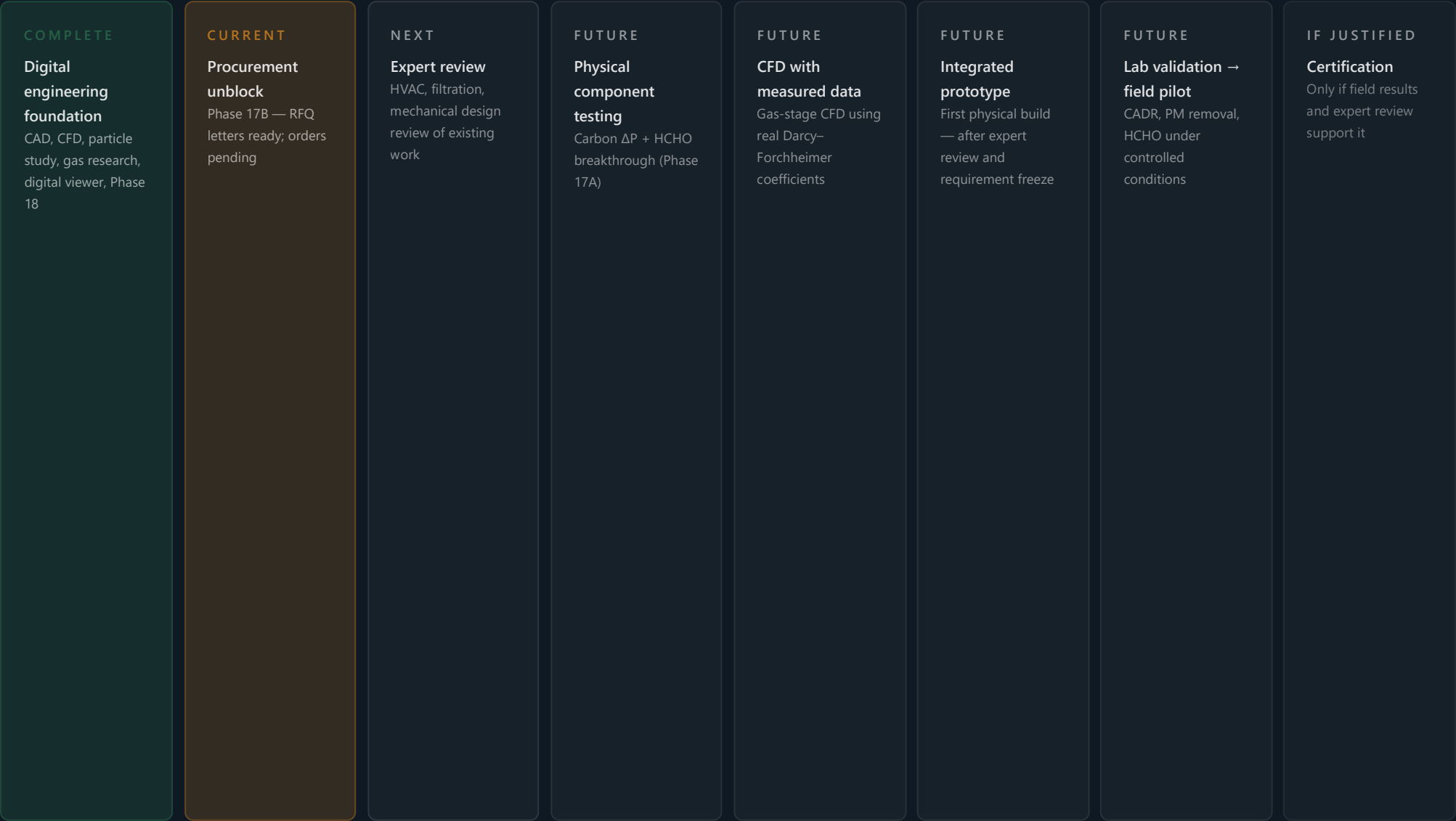
Manufacturing

Material sourcing, tolerances, cost estimation, supply chain

The ten questions we most want answered by experienced engineering reviewers.

- | | |
|--|--|
| <p>01 Is the cross-flow tower architecture sensible for the intended deployment, or does the turning geometry impose fundamental disadvantages?</p> <hr/> | <p>06 What test standards are relevant for validating this class of air-treatment device in India? What is the certification path?</p> <hr/> |
| <p>02 Is the Systemair KVO 250 the appropriate fan class? What would a professional fan selection look like for this system resistance?</p> <hr/> | <p>07 What materials and safety requirements have been overlooked in the current component selections?</p> <hr/> |
| <p>03 How should the HEPA filter be mounted and sealed in a manufacturable enclosure to prevent real-world bypass?</p> <hr/> | <p>08 What should be simplified, removed, or deferred before a first physical prototype is attempted?</p> <hr/> |
| <p>04 What is a realistic pressure-drop budget across all treatment stages, and how much fan headroom is genuinely available?</p> <hr/> | <p>09 Are the RANS CFD simulations useful as a starting point, or do specific aspects — turbulence model, wall treatment, boundary conditions — need to be revisited?</p> <hr/> |
| <p>05 How should the carbon media bed be sized — area, depth, packing — and where should it sit in the treatment sequence?</p> <hr/> | <p>10 What would make this system maintainable in real operation — filter replacement intervals, carbon bed life, cleaning indicators, serviceable access points?</p> <hr/> |

Where this project is, and where it could go.



Roadmap is indicative — not a committed timeline. Progress in each stage depends on resources, expertise, and results of prior stages.

Digital engineering is much further developed than physical validation — intentionally.



No fake overall percentage

Digital and physical maturity are fundamentally different — one can be high while the other is zero. The honest position: digital engineering foundation is strong; physical evidence is zero.

This is the expected sequence

A simulation-first approach deliberately produces this maturity profile. Physical validation comes after design decisions are well enough informed to make testing purposeful.

The result is not a finished purifier. The result is a better-defined engineering problem.

Before this work, the AQI Tower was a visual concept — a plausible shape with no engineering basis for any of its parameters, filter choices, fan selection, treatment stages, or performance claims.

After this work, the team knows: why a particular geometry produces less pressure loss; where bypass was hiding; what the fan curve looks like; what particle trajectory data can and cannot tell you; which gas is the right research target; and what experiment is needed before gas-stage CFD can be trusted.

None of these answers existed before the work began. All of them reduce the cost and risk of physical development.

Quantified improvements from simulation

5×

pressure loss reduction
V01 → GEO_C

45%

bypass discovered
and sealed in model

4.7×

riser entry CoV
improvement

Decisions that saved future effort

Water spray removed: Eliminated 12+ components, Legionella risk, pump energy, and 35–125 Pa ΔP before any hardware was purchased.

Gas CFD blocked: Prevented a simulation with invented coefficients that would have looked credible but been meaningless.

Questions now precisely defined

Every remaining uncertainty is now an answerable question with a known experimental method. That is not the same as having the answer — but it is far more useful than not knowing what to ask.

Responsible engineering means being equally clear about what you do not know.

No validated CADR

Clean Air Delivery Rate requires physical measurement under standard conditions. No CADR value has been established for the AQI Tower at any stage of development.

No PM2.5 removal efficiency

The H13 classification is a component rating at test conditions. Without $\eta(d_p)$ at operating velocity, no system-level PM2.5 efficiency can be calculated from available data.

No HCHO removal efficiency

No breakthrough experiment has been performed. The carbon stage has no measured capacity, no breakthrough time, and no gas-phase CADR.

No claim of physical bypass = 0%

CFD bypass $\approx$ 0% is a modelled result. Whether a fabricated tower is physically airtight depends entirely on manufacturing quality — not CFD.

62% is not particle removal

62% is a simulated particle arrival fraction at the filter plane. Filter capture efficiency $\eta(d_p)$ has not been applied — it is not available for this product.

1,332 m³/h is not measured airflow

1,332 m³/h is a RANS CFD result from a partially converged case. It is a useful screening estimate. Real tower airflow has never been measured.

KEY SIMULATION RESULTS AT A GLANCE

ALL VALUES BELOW ARE SIMULATION ONLY — NOT PHYSICALLY VALIDATED — PARTIALLY CONVERGED RANS

1,496

M³/H

GEO_C empty tower airflow

SIMULATED

3.5 Pa

FAN HEAD

GEO_C empty (no filters)

SIMULATED

1,332

M³/H

with H13 HEPA (CASE_M_SEALED)

SIMULATED

111 Pa

FILTER ΔP

CASE_M_SEALED

SIMULATED

45%

BYPASS (BEFORE FIX)

Discovered Phase 10A

SIMULATED

~0%

BYPASS (AFTER FIX)

bypassSeal — modelled only

SIMULATED

62.1%

PARTICLE ARRIVAL

0.3–2.5 μm at filter plane

≠ CAPTURE

688 Pa

FAN SHUTOFF HEAD

KVO 250 ±20 Pa (catalogue)

MFR DATA

≥99.95%

FILTER EFFICIENCY

H13 EN 1822 at MPPS only

MFR CLASS.

63,780

CFD CELLS

GEO_C mesh

SIMULATED

100 μg/m³

HCHO BASELINE

Planned challenge concentration

NOT TESTED

0

PHYSICAL PROTOTYPES

No hardware built or tested

FUTURE WORK

CLOSING

An idea became a model. The model became questions. The questions became engineering.

The next stage is to bring experienced people, real hardware, and physical evidence into the loop — to test what simulation suggested, verify what literature predicted, and correct what both got wrong.

The AQI Tower team does not claim to have designed a finished product. They claim to have done honest exploratory engineering on a difficult problem — and to have built a foundation that makes the next stage of work more purposeful than it would have been without it.

18+

documented phases

10

CFD simulation cases

0

fabricated results

∞

questions answered honestly

Selected key values and their direct repository sources

Airflow CFD

GEO_C airflow 1,496.055 m³/h — *results/GEOMETRY_OPTIMIZATION/GEO_C/metrics.json: fan.Q_m3_h*

GEO_C fan head 3.500 Pa — *metrics.json: fan.curve_static_Pa*

V01 pressure loss 17.326 Pa — *results/GEOMETRY_OPTIMIZATION/comparison_metrics.json:*

V01.tower_total_pressure_loss_Pa

V01 riser reverse 23.12% — *comparison_metrics.json: V01.riser_reverse_percent*

GEO_C cells 63,780 — *GEO_C/metrics.json: cell_count*

GEO_C riser entry CoV 0.234 — *metrics.json: sections.riser_entry_above_turn.normal_velocity_CoV*

Filter and sealed case

CASE_M_SEALED airflow 1,332.205 m³/h — *results/FILTER_H13/bypass_analysis_sealed.json:*

CASE_M_SEALED.total_flow_m3_h

Filter ΔP 111.181 Pa — *bypass_analysis_sealed.json: CASE_M_SEALED.filter_delta_P_area_mean_Pa*

HEPA face velocity 1.0525 m/s — *bypass_analysis_sealed.json: filter_face_velocity_m_s.mean*

Bypass before 45.05% — *bypass_analysis_sealed.json: comparison.CASE_M_bypass_fraction*

Bypass after 9.997×10^{-10} — *bypass_analysis_sealed.json: CASE_M_SEALED.bypass_fraction*

Particle tracking

All CFD results: SIMULATION ONLY — PARTIALLY CONVERGED RANS — NOT PHYSICALLY VALIDATED. Report date: September 2026. Repository: D:/DUDE/AQI-TOWER

0.3 μm arrival 62.1% (139/224) — *results/PARTICLE_CFD/particle_arrival_fractions.json: results.0p3um*

10 μm arrival 61.2% (137/224) — *particle_arrival_fractions.json: results.10um*

Wall deposit ~37.9% — *particle_arrival_fractions.json: results.0p3um.wall_deposit_fraction*

Fan and filter (manufacturer data)

KVO 250 shutoff ~688 Pa ±20 Pa — *docs/FAN_CURVE_AUDIT.md, Phase 6D PyMuPDF extraction, Systemair catalogue*

Freudenberg SF13-B H13 — *Manufacturer product page, Phase 6C confirmation*

HEPA rated ΔP 300 Pa at 2.84 m/s — *Freudenberg manufacturer page (single point)*

Phase 18

NO-GO decision — *docs/PHASE18_WATER_SPRAY_FEASIBILITY.md, Section 15*

Greenfield gap — *Hinds (1999) Aerosol Technology 2nd ed.; Seinfeld & Pandis (2016)*

Legionella risk standard — *ASHRAE 188-2021; CIBSE TM13 (2000)*

Procurement (confirmed India prices, 2026-09-07)

SDP810 ₹4,748 — *Tanotis India, confirmed online listing*

Rotameter A ₹1,950 — *Shree Aashapura Mumbai, IndiaMART listing*